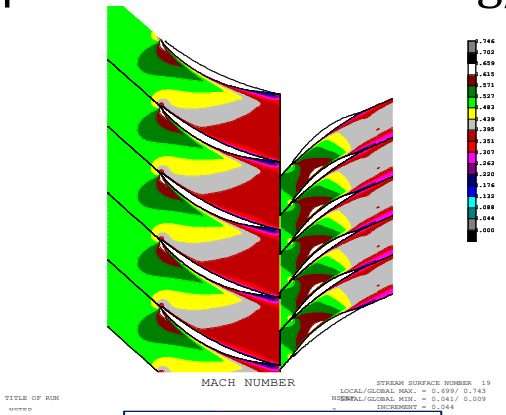
M2794.003800 Flow and Design

Blade Optimization :
Single-Stage Axial Flow Compressors

Term Project 2026-1 | 김건오

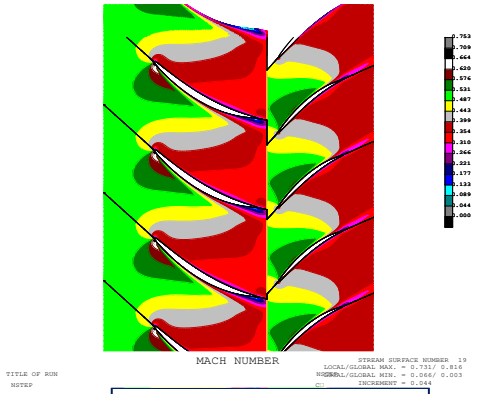
Abstract

- Compressor A: $\dot{m} = 40$ kg/s, $r_t = 0.40$ m, $b_r = 0.040$ m, $b_t = 0.032$ m



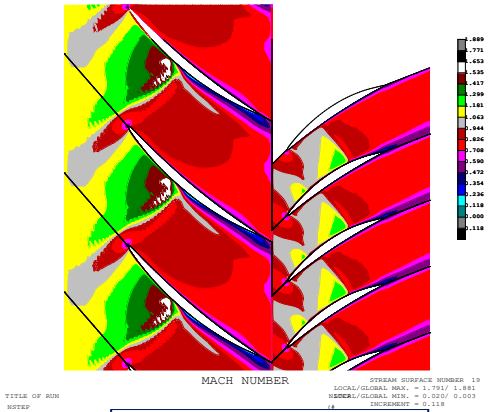
$\eta_{tt} = 0.9123$

$Z_w = 0.56$
EXPO $= 1.0$
Stator:
TKMAX $= 0.075$
XTKMAX $= 0.400$



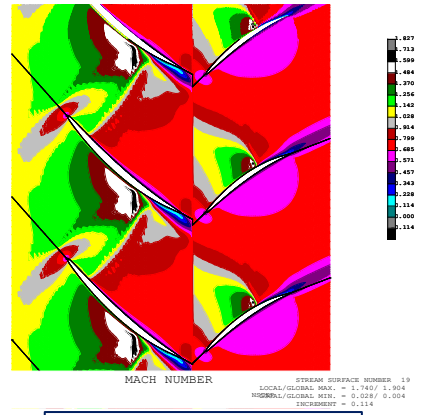
$\eta_{tt} = 0.9356$

- Compressor B: $\dot{m} = 80$ kg/s, $r_t = 0.75$ m, $b_r = 0.075$ m, $b_t = 0.060$ m



$\eta_{tt} = 0.7917$

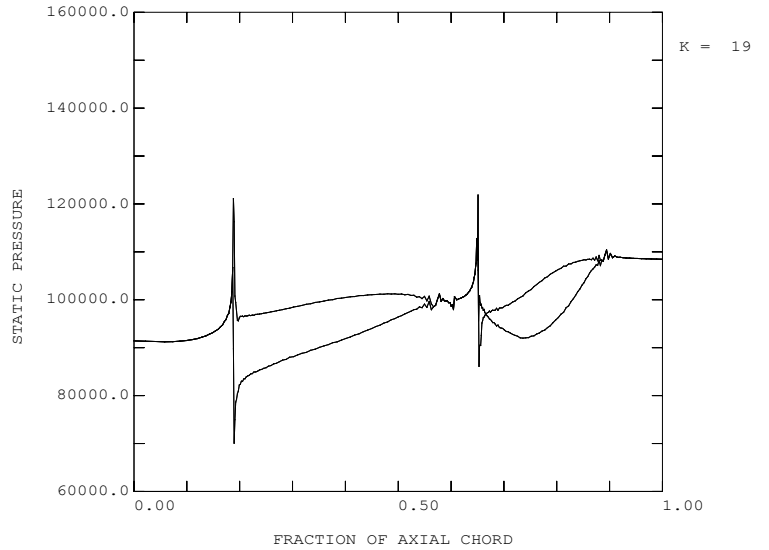
$Z_w = 0.56$
EXPO $= 1.0$
Stator
TKMAX $= 0.075$
XTKMAX $= 0.400$

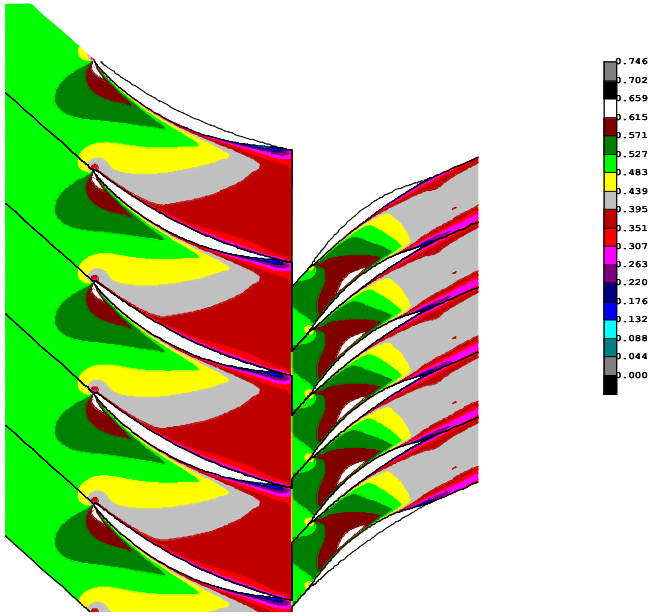


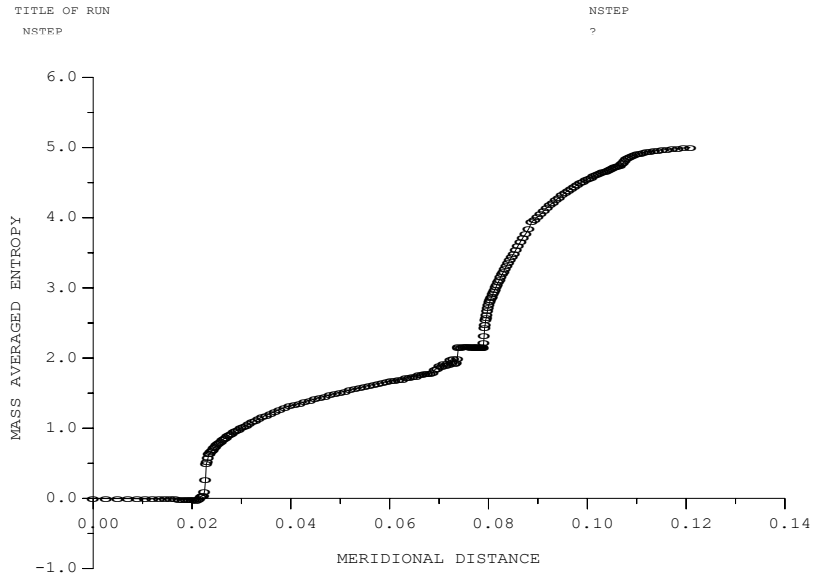
$\eta_{tt} = 0.8111$

Base case – Compressor A

- Larger entropy rise at **Stator**
  - Blade decrease → skin friction
  - Thinner stator blade ≈ rotor
  - More turning at the front blade

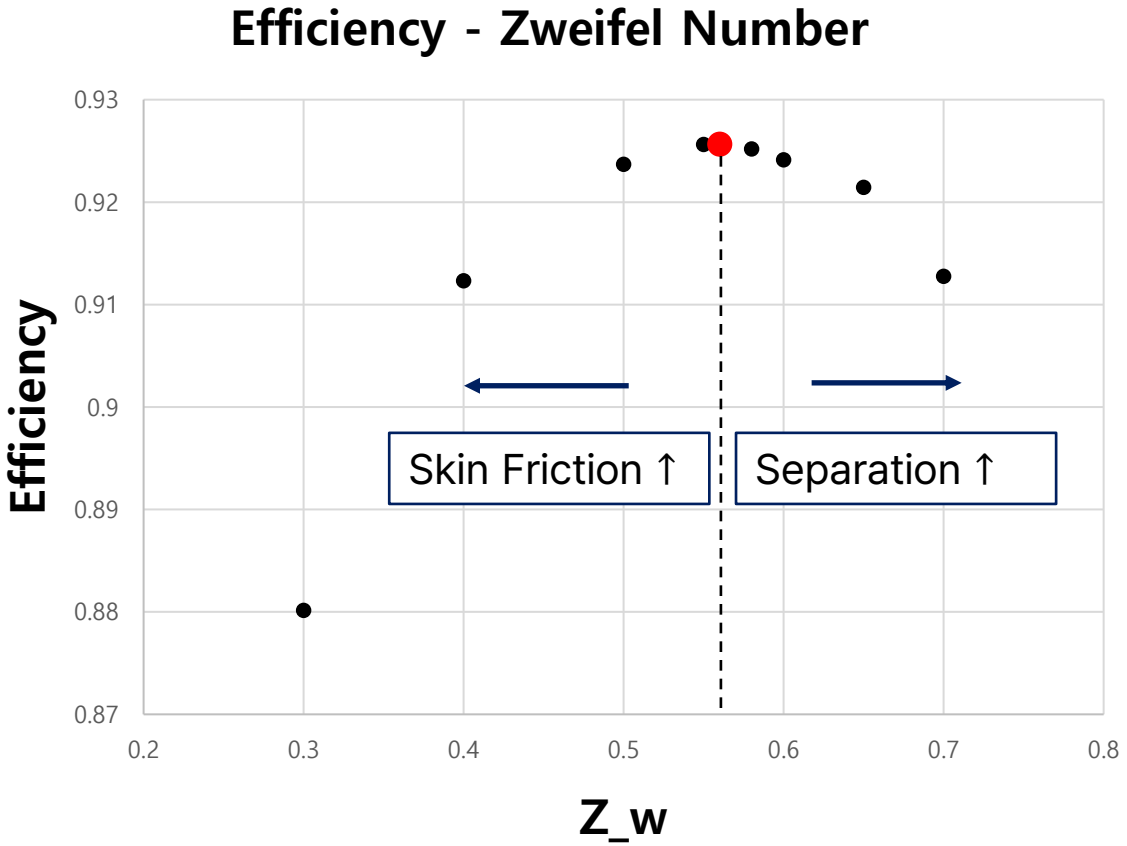




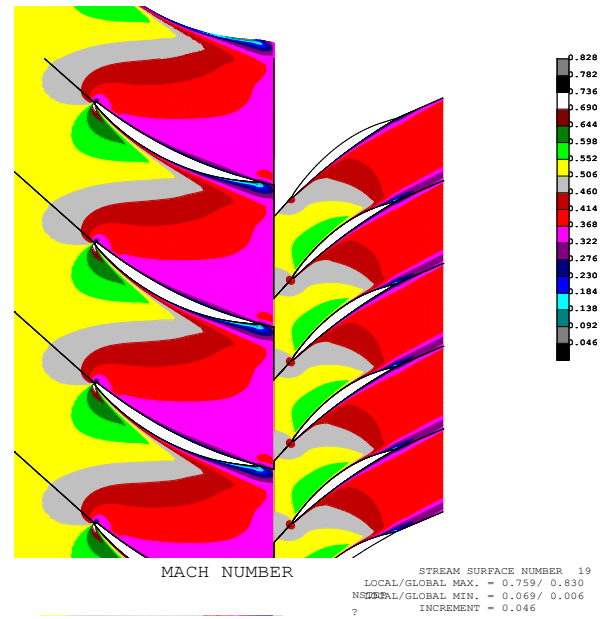


Optimization – Zweifel Number

- Max $\boldsymbol{\eta_{tt} = 0.9256}$ at $Z_w = 0.56$, rotor blades 71 → 51
- $Z_w = 2(s/b)\cos^2\alpha_2\,(\tan\alpha_1 + \tan\alpha_2)$

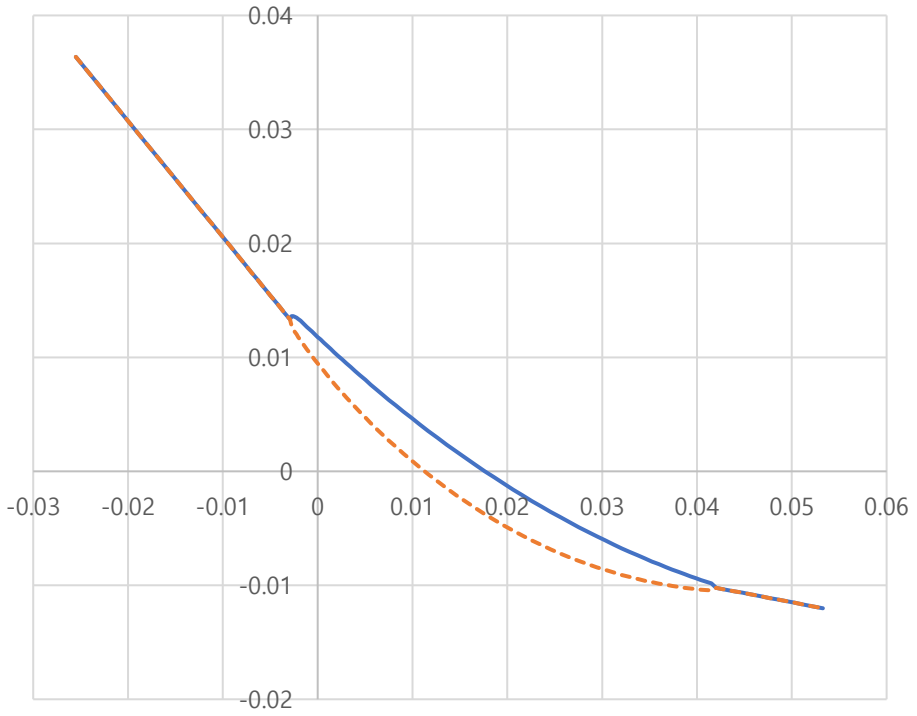


$$\mathrm{DF} = \left(1 - \frac{w_2}{w_1}\right) + \frac{\Delta c_\theta}{2w_1}\frac{s}{b}$$

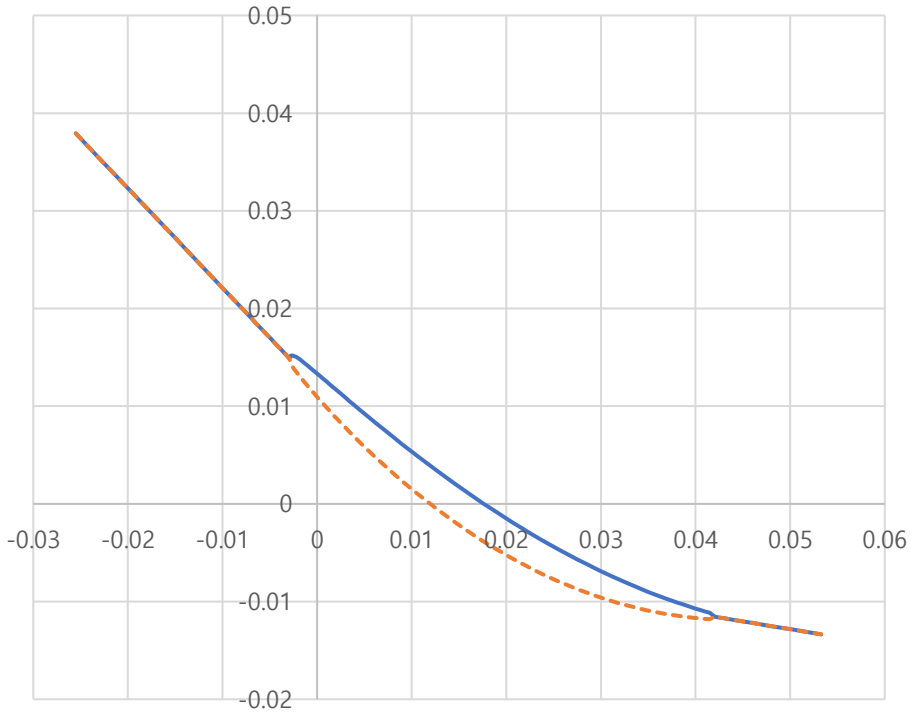


Optimization – Blade Shape, Rotor

- Stator: $\mathrm{TKMAX}, \mathrm{XTKMAX} = 0.1, 0.45 \rightarrow \mathbf{0.075}, \mathbf{0.4}$
- Stator & Rotor: $\mathrm{EXPO} = 1.35 \rightarrow \mathbf{1.0}$



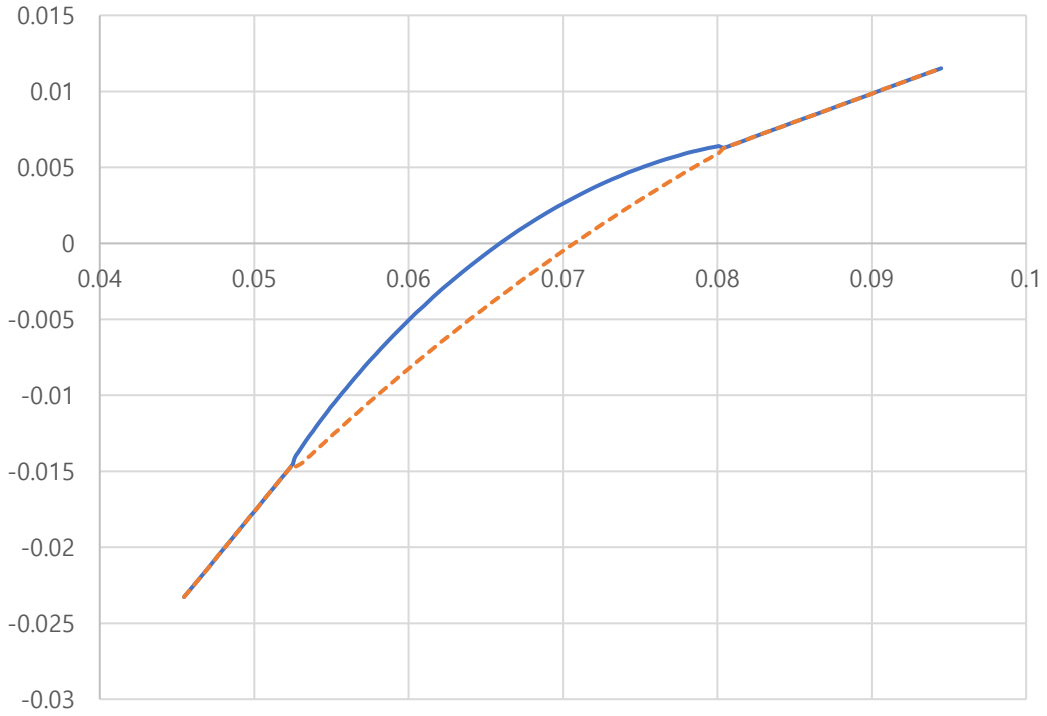
Base Rotor



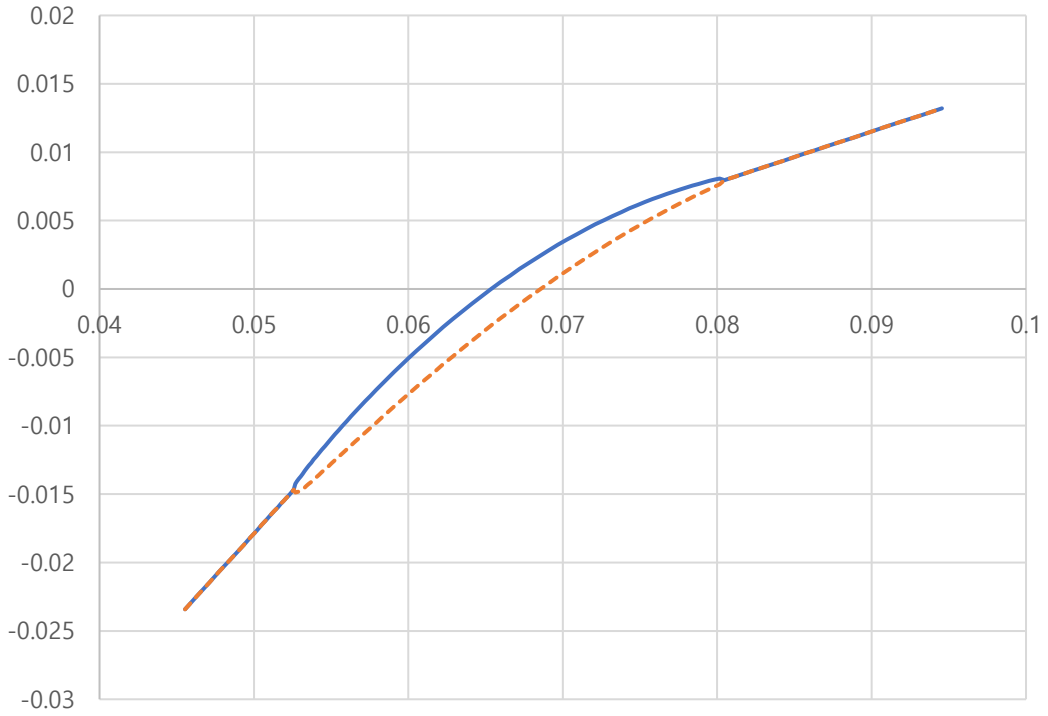
Optimized Rotor

Optimization – Blade Shape, Stator

- Stator: $\mathrm{TKMAX} = 0.1, \mathrm{XTKMAX} = 0.45$ → $\mathbf{TKMAX = 0.075, XTKMAX = 0.4}$
- Stator & Rotor: $\mathrm{EXPO} = 1.35$ → $\mathbf{EXPO = 1.0}$



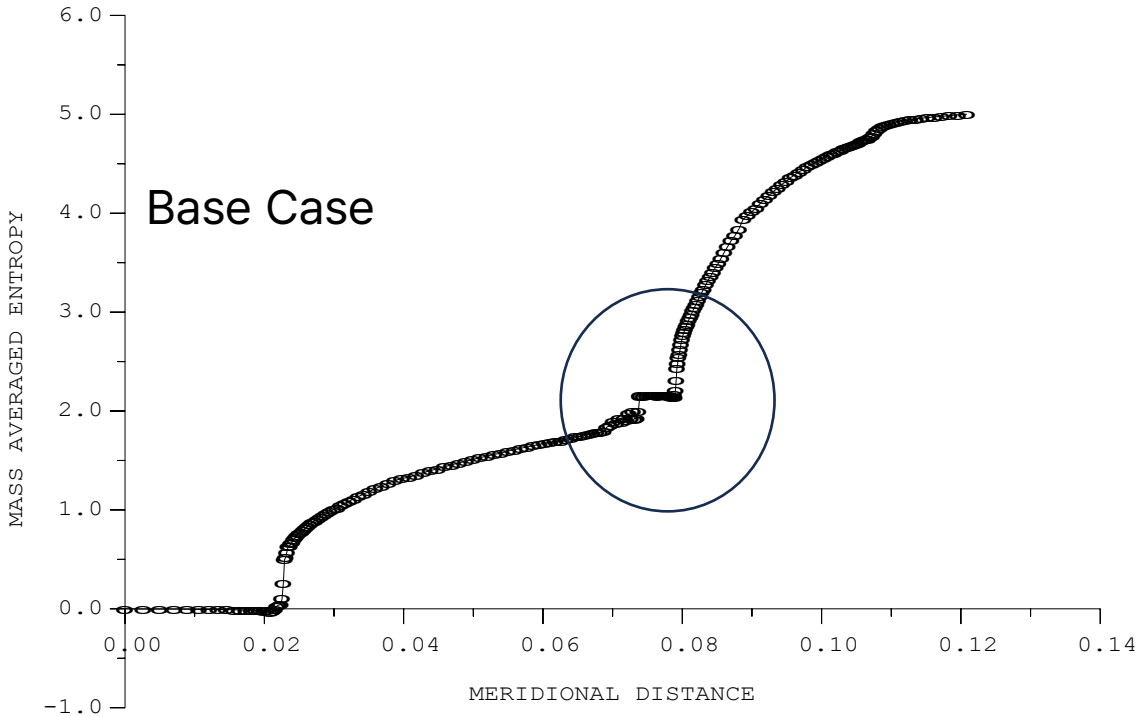
Base Stator

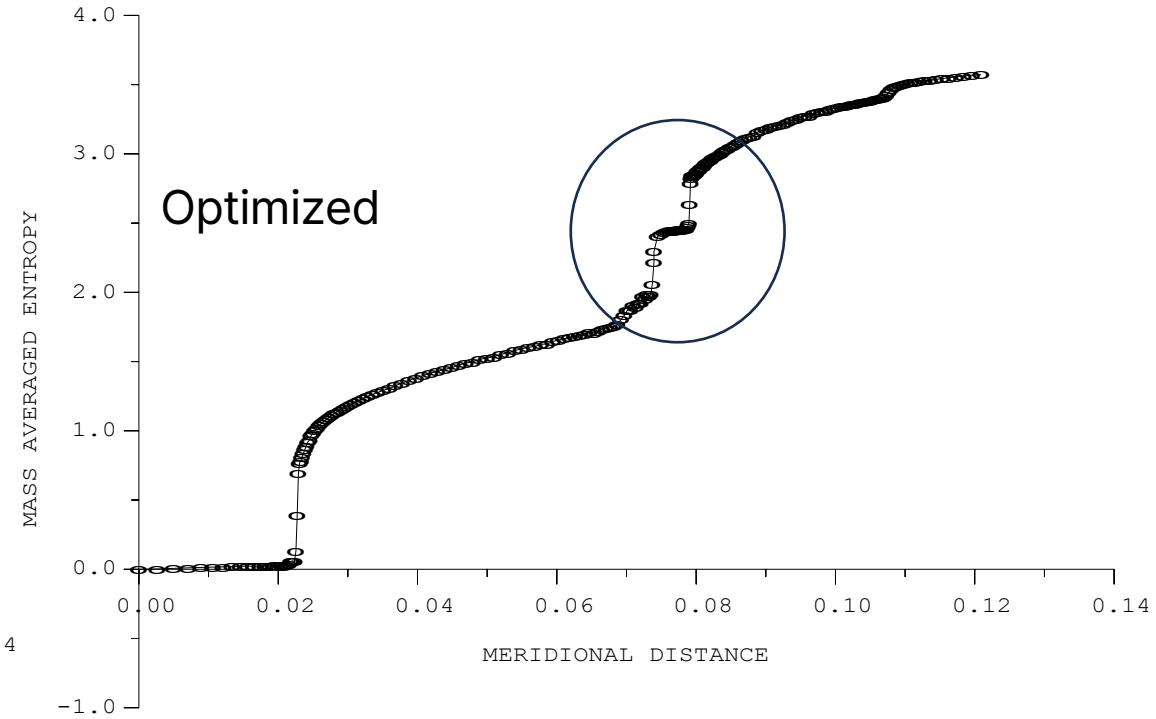


Optimized Stator

Optimization, Blade Shape

- Thinner stator blade
- EXPO $1.35 \rightarrow 1$, Turning at the rear → Circular arc
- $\boldsymbol{\eta_{tt} = 0.9276}$



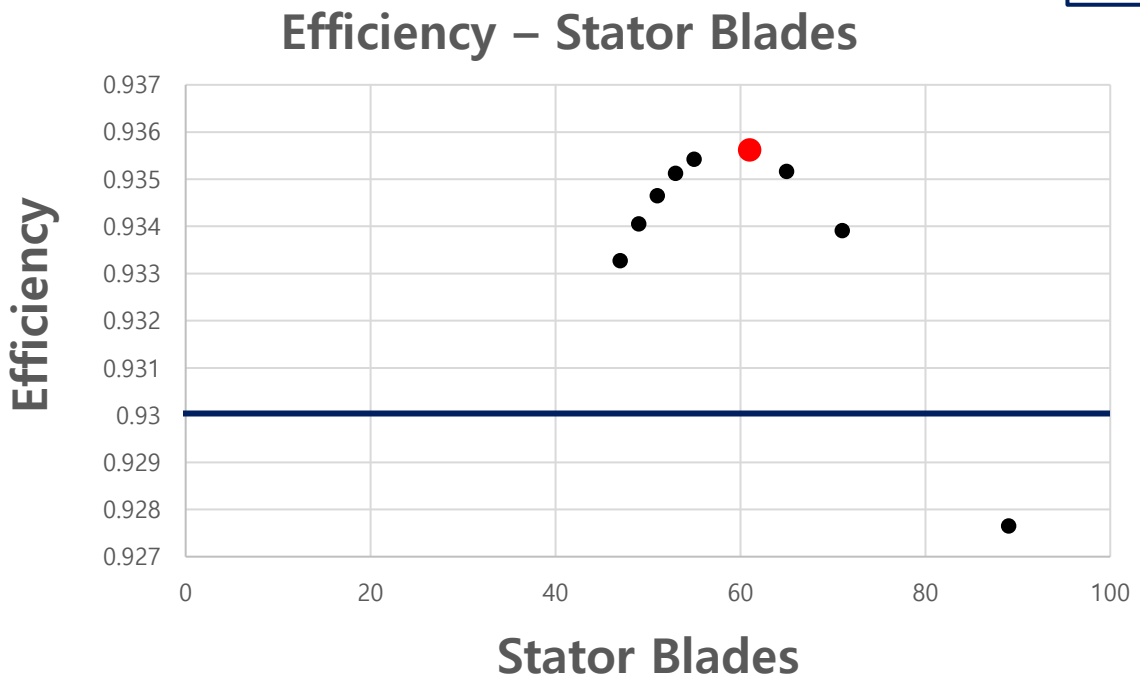


Stator Blade Number Control

- $Z_w = 0.4 \rightarrow 0.56$
  - Rotor blades: 71 → 51
  - Stator blades: 125 → 89
- **Reduce More Stator Blades (stagen.dat)**

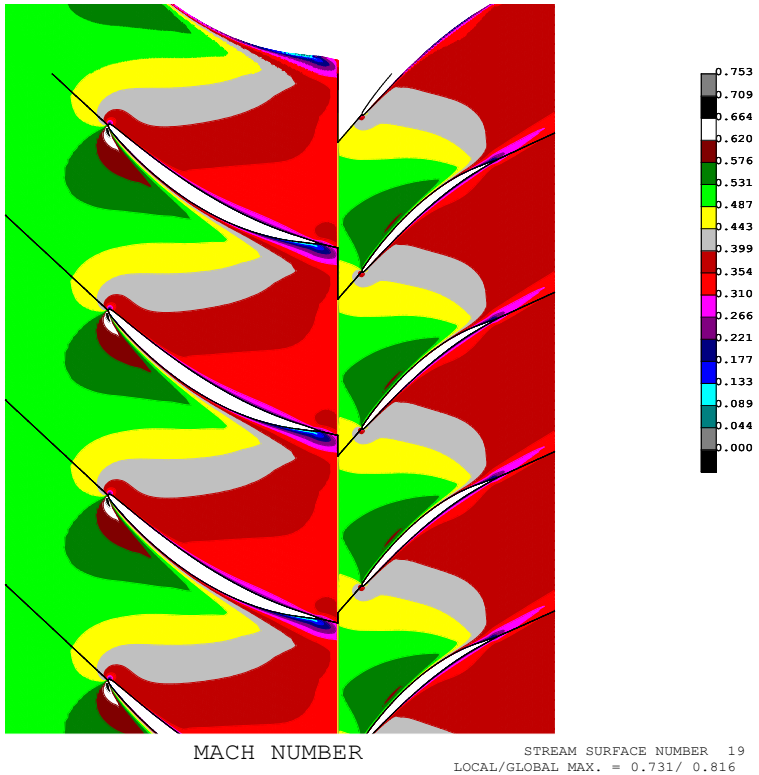
Blade Number Trade-off
- Many, skin friction increase
- Few, Loading per blade rise

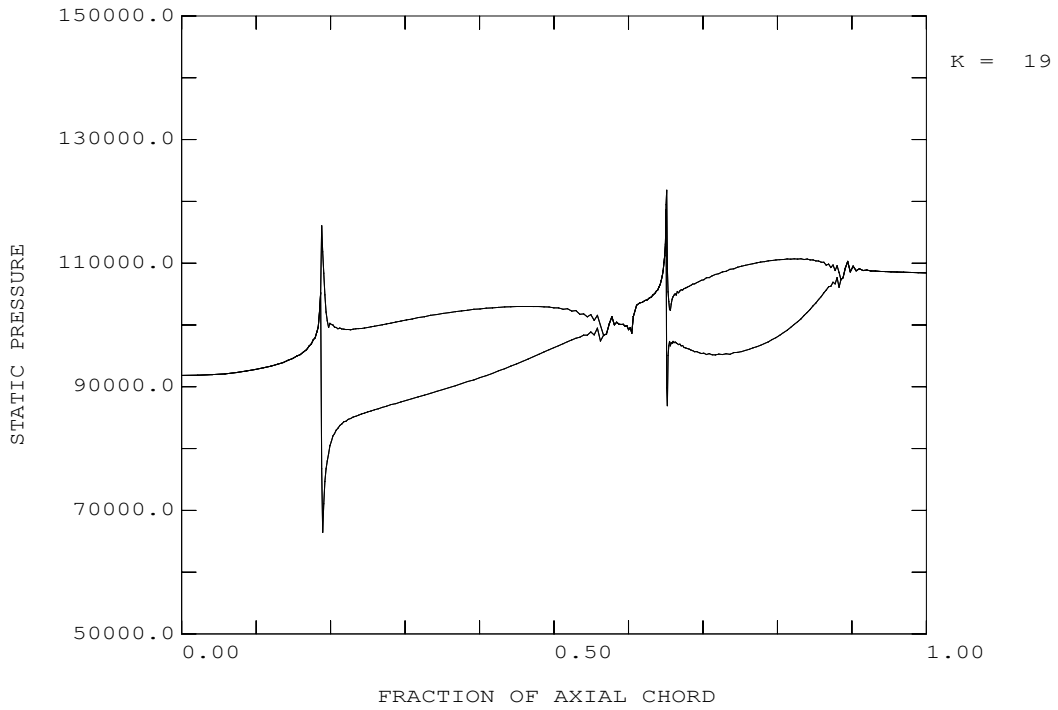
→ Separation risk rises



Final Optimization

- $Z_w = 0.56$, $\mathrm{EXPO} = 1$, Stator $\mathrm{TKMAX} = 0.075, \mathrm{XTKMAX} = 0.4$, blade 61
- Stator blade 61 → $\boldsymbol{Z_w \approx 0.817}$
- $\boldsymbol{\eta_{tt} = 0.9356}$**, 2.56% increased**

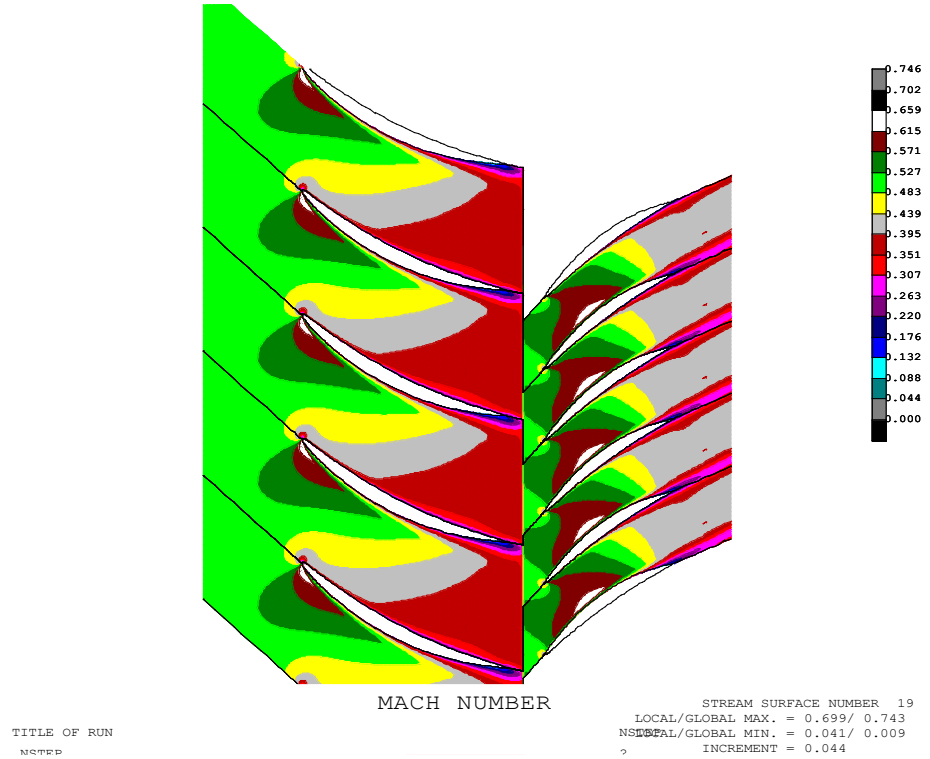




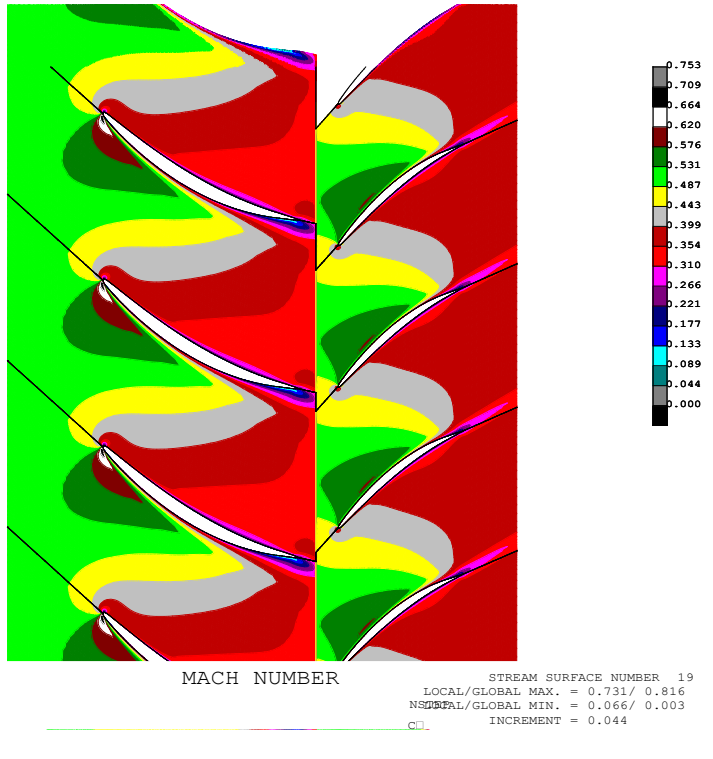
Comparative Analysis –Mach Contour

- Optimized → lower max Mach Number at stator (0.659→0.620)
- **Loading shifted forward** → smoother pressure recovery
- Thinner blade, spacing increased → lower skin friction loss

Base Case



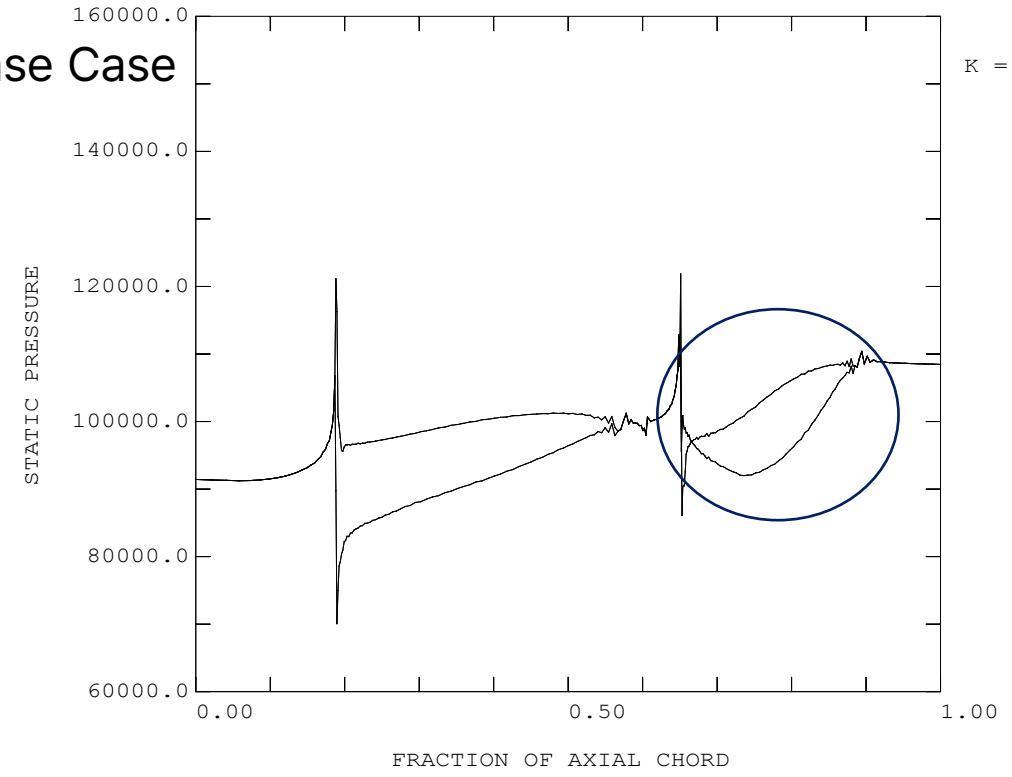
Optimized



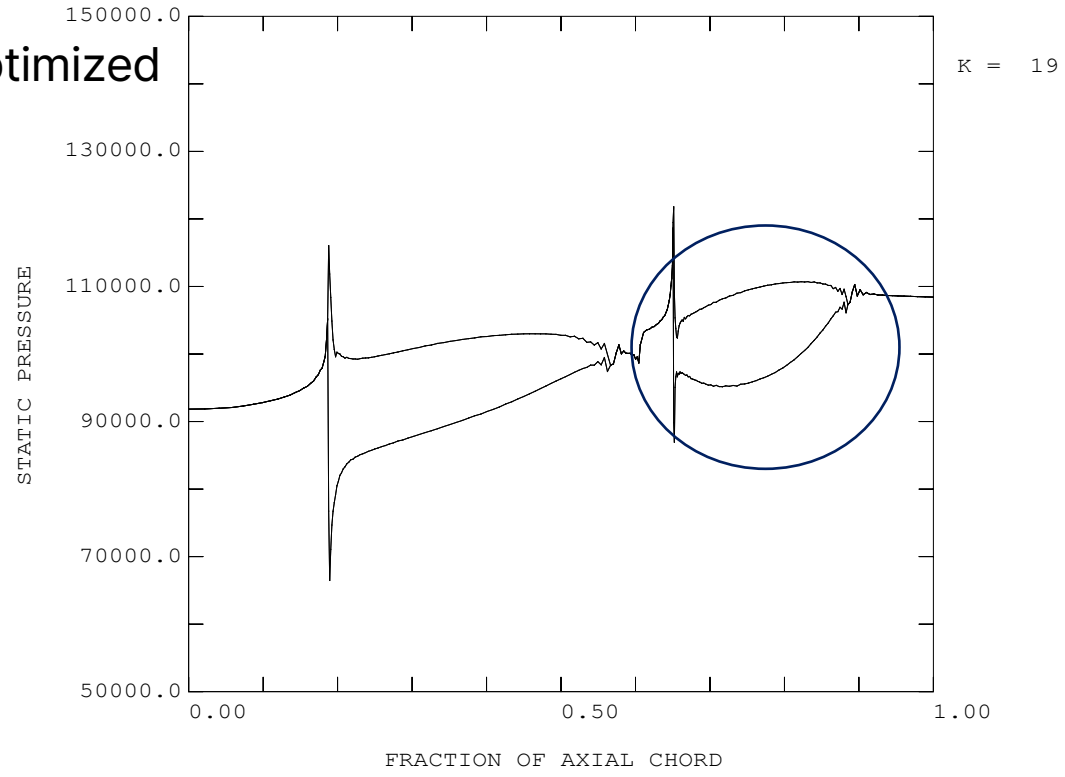
Comparative Analysis – Surface Pressure Distribution

- Stator suction surface → **Milder adverse pressure gradient**

Base Case

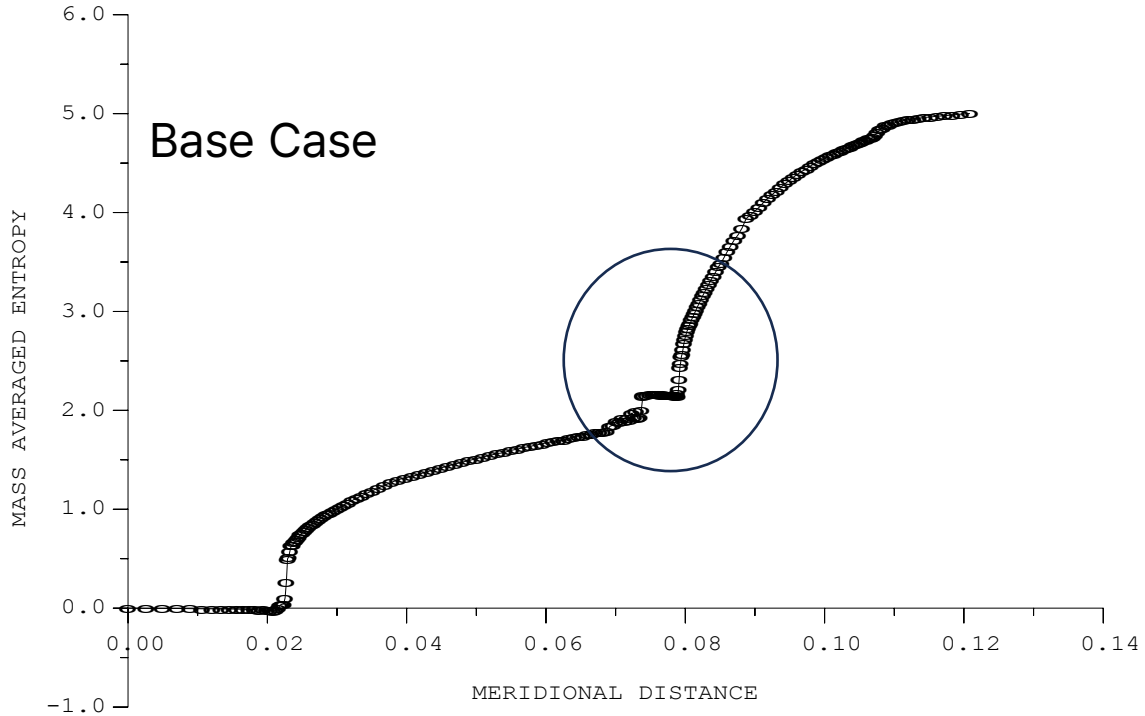


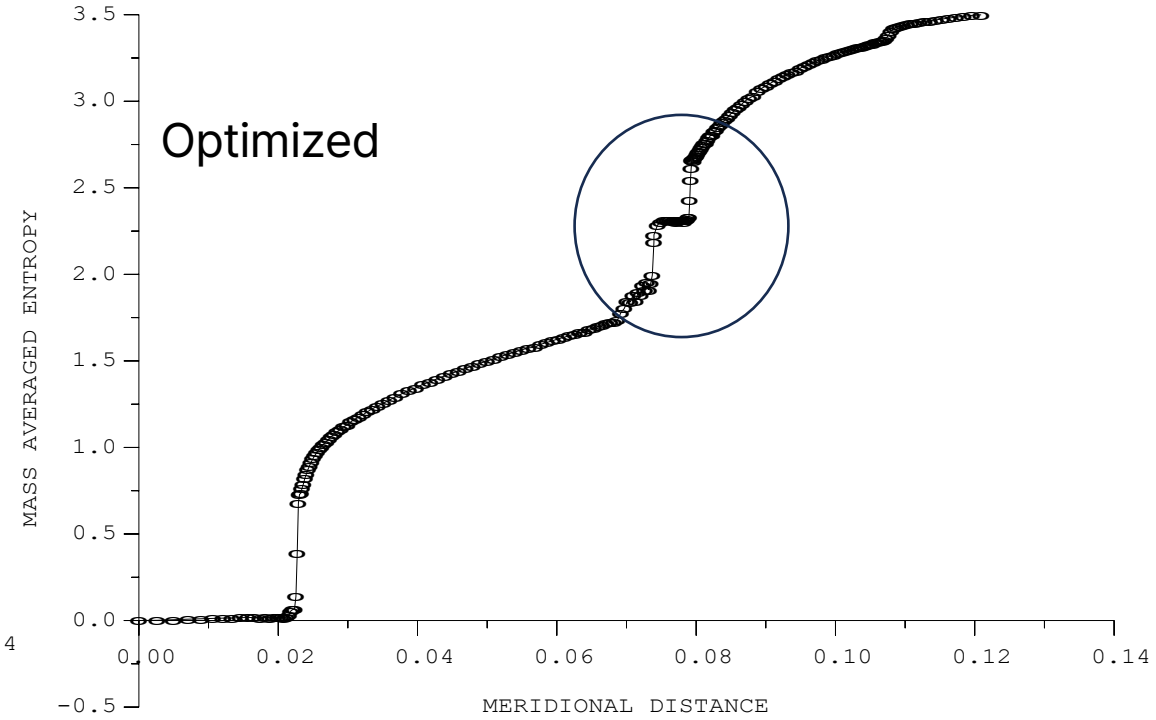
Optimized



Comparative Analysis – Entropy Rise

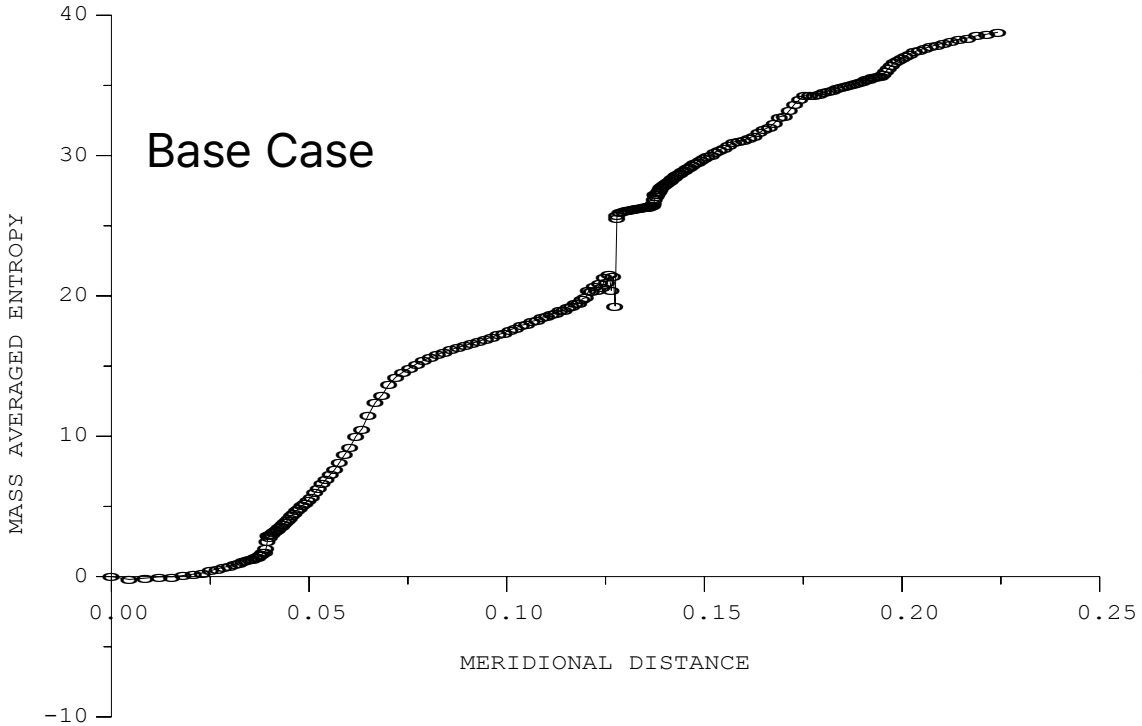
- Total Entropy change decreased (5.0->3.5), loss decreased
- ΔS_r similar, ΔS_s mainly reduced
- ΔS focused at LE of the blade

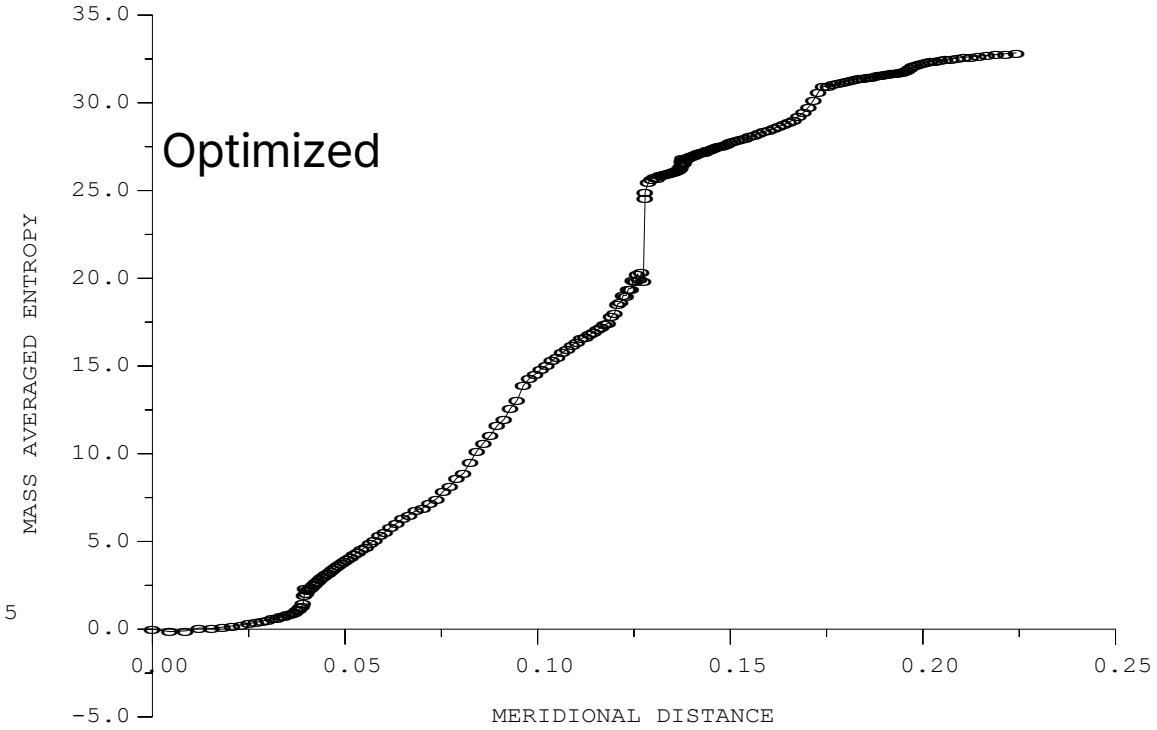




Optimization Design for Compressor B

- $Z_w = 0.56$, $\mathrm{EXPO} = 1$, Stator $\mathrm{TKMAX} = 0.075, \mathrm{XTKMAX} = 0.4$
- Rotor Blade: 43, Stator Blade: 80 → 45
- $\boldsymbol{\eta_{tt} = 0.8111}$**, 2.44% increased**

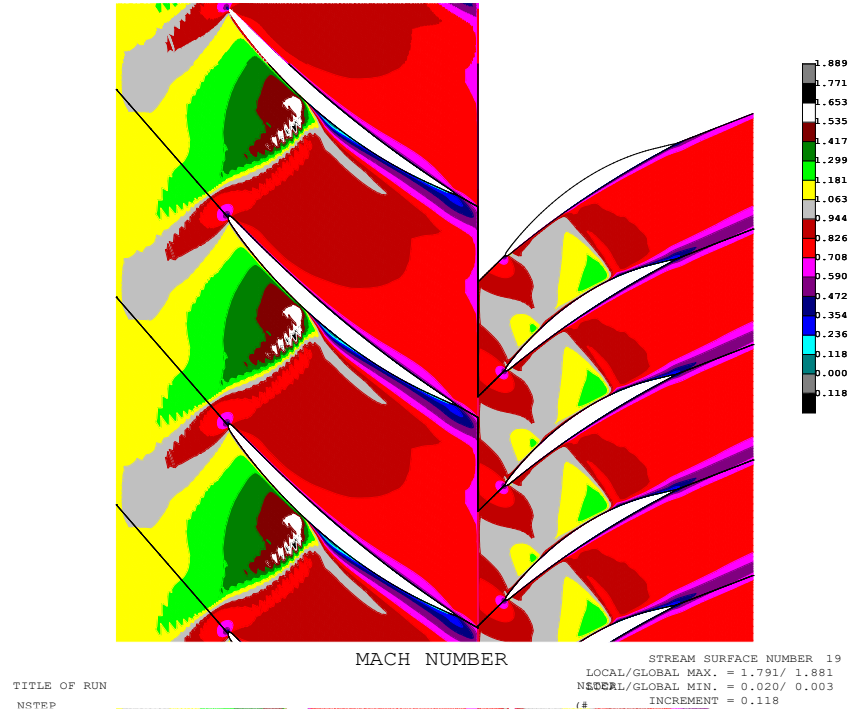




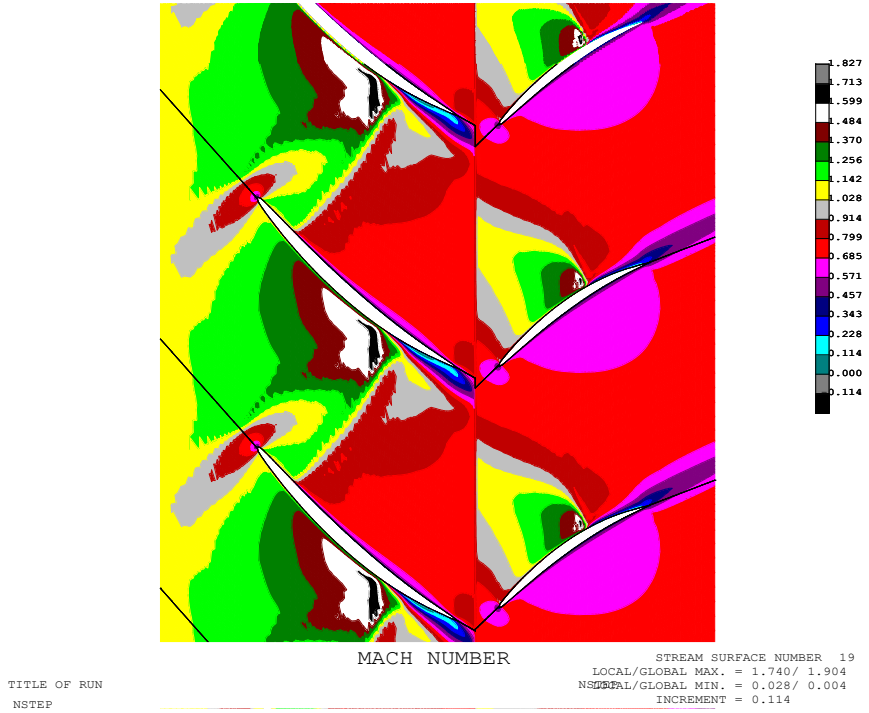
Compressor B – Mach Contour

- Thinner blades, larger spacing → **Compressor B also improved**
- Ma>1 region remains → Shock loss, efficiency lower than compressor A
- **Shock is at LE** for optimized stator → Peak efficiency

Base Case



Optimized

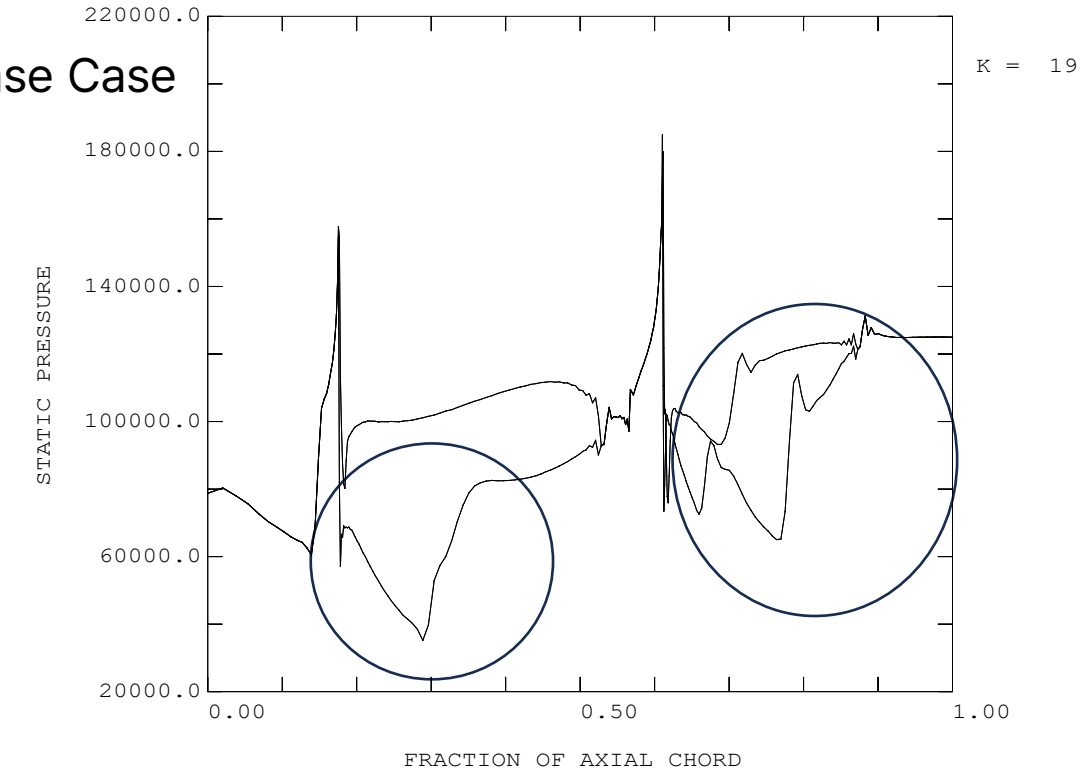


Cesare Hall and S. Larry Dixon, Fluid Mechanics and Thermodynamics of Turbomachinery Seventh Edition (2014)

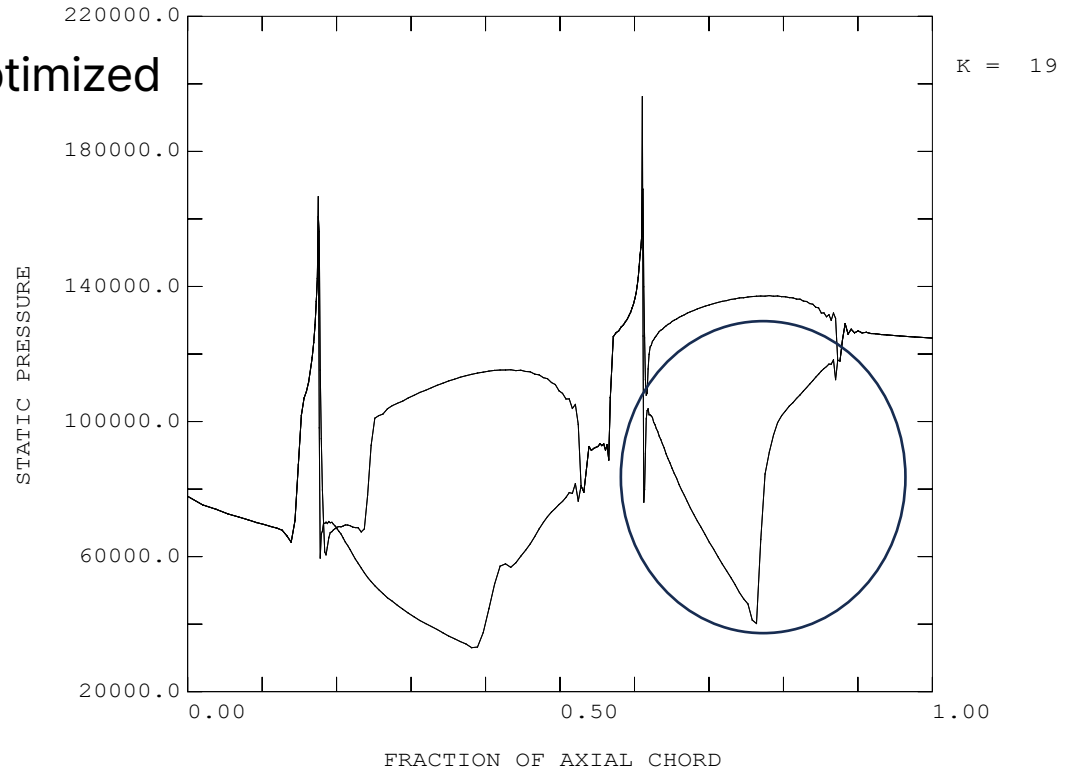
Compressor B – Pressure Distribution

- Stator SS, one peak at optimized geometry, **two peak at base case**
- Rotor SS, Large adverse pressure gradient at base case

Base Case



Optimized

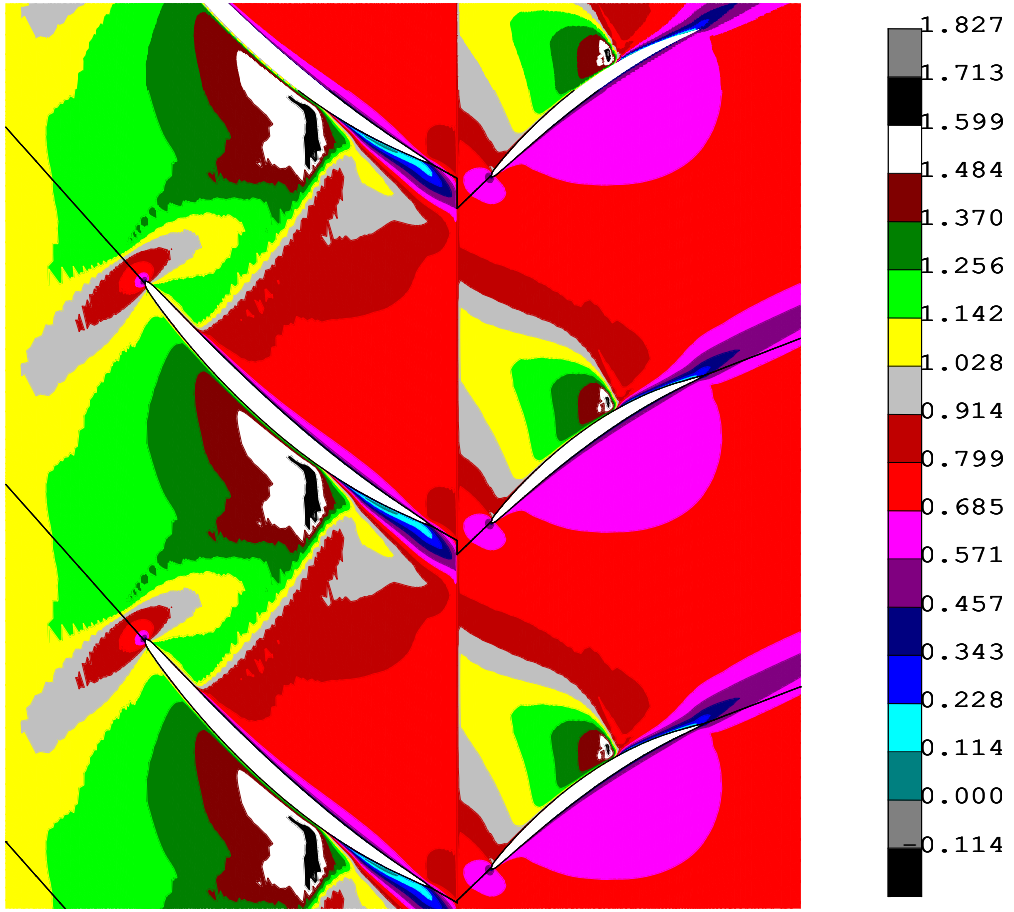


Compressor B Efficiency Limitation

- Large Tip Speed $U_t = \left(5000 \times \frac{\pi}{30}\frac{\text{rad}}{\text{s}}\right) \times 0.75\ \text{m} = 392.7\ \text{m/s}$

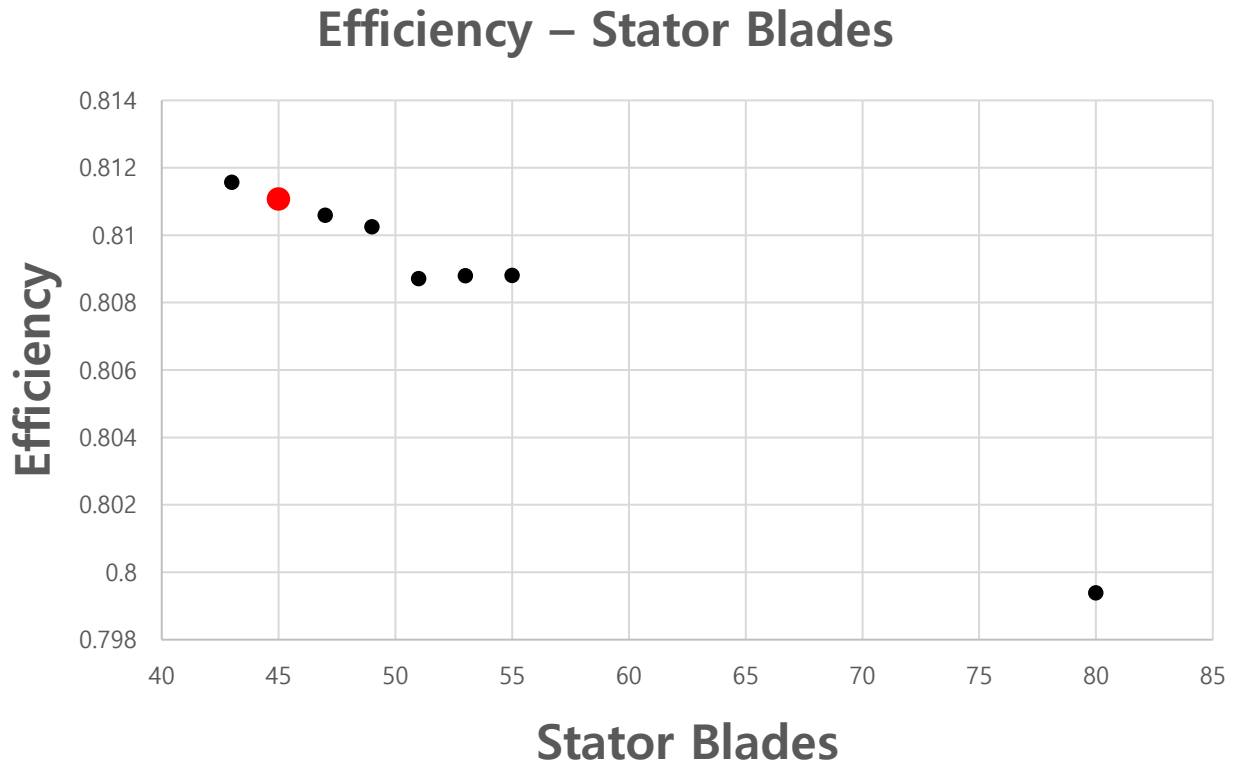
Compressor B – Shock Loss
- Mach up to ~1.8
- Shock + B.L. interaction

→ Loss rises



Blade Resonance Risk

- Maintaining **coprime** blade numbers
  - Rotor wakes strike stator blades with staggered timing
  - Reducing the risk of resonance and high-cycle fatigue



Carlos Perez Arroyo et al., Towards the Large-Eddy Simulation of a full engine: Integration of a 360 azimuthal degrees fan, compressor and combustion chamber. Part I: Methodology and initialization (2021)

Conclusions

- Adjusting Z_w, EXPO, Stator $\mathrm{TKMAX}, \mathrm{XTKMAX}$
- **Larger spacing, Smoother turning, Thinner blade**
  - Compressor A: $\eta_{tt} = 0.9123 \rightarrow 0.9356$
  - Compressor B: $\eta_{tt} = 0.7917 \rightarrow 0.8111$
- Optimization strategy works for compressor B, but η_{tt} limited
  - Shock-related loss occurs due to high tip speed
  - Shock location is also important for efficiency

Q&A